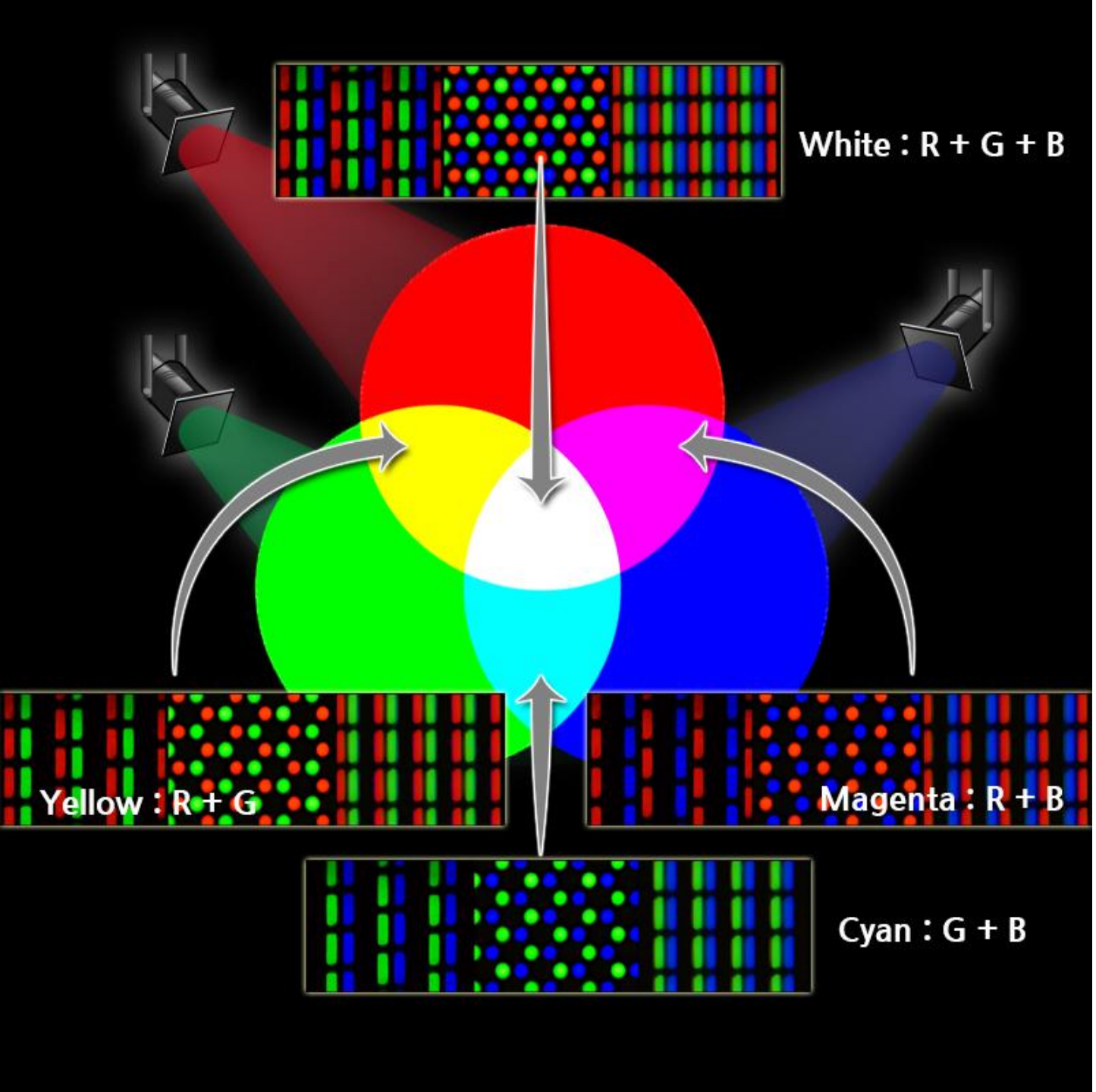
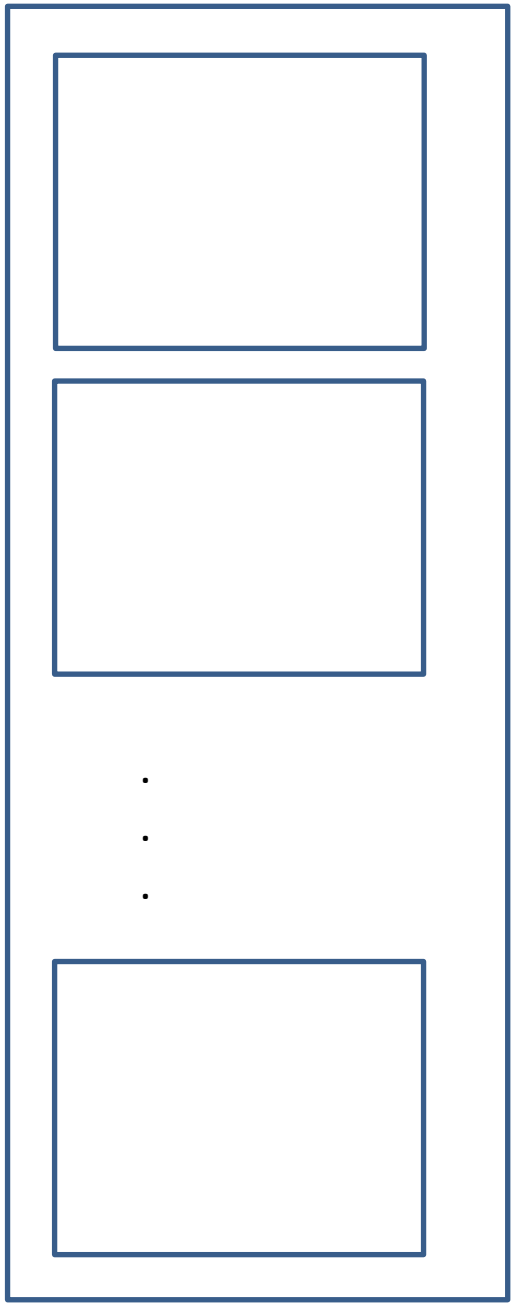
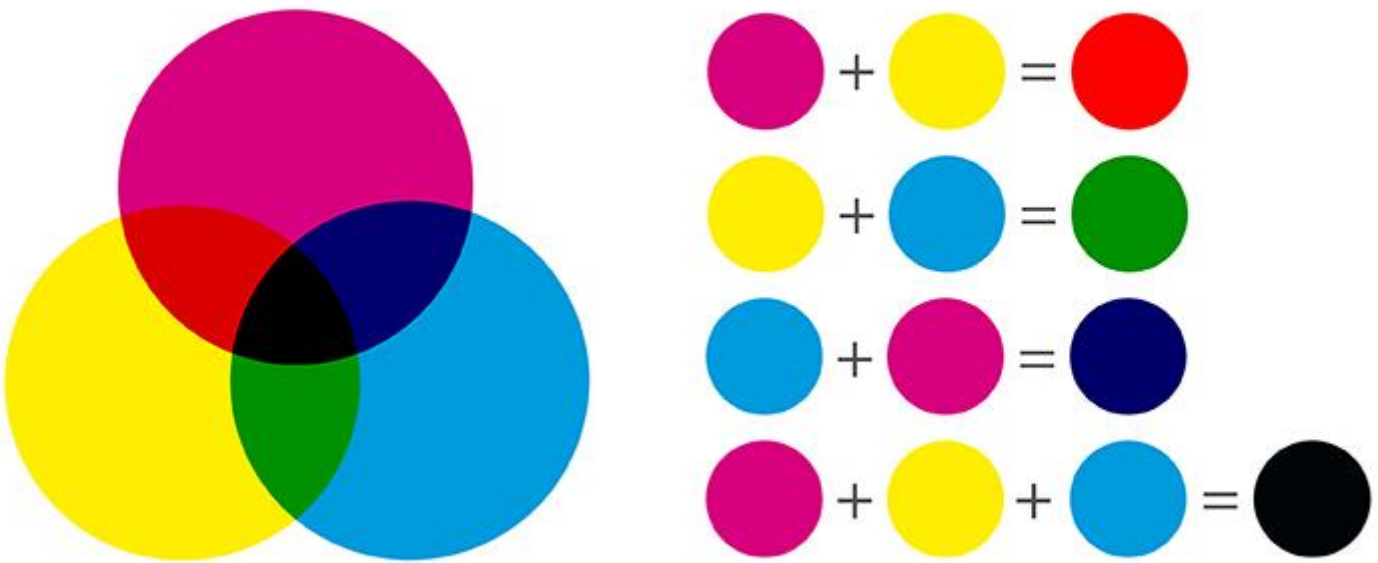

White : R + G + B
Yellow : R + G
Magenta : R + B
Cyan : G + B



CMYK



DEEP LEARNING

CONVOLUTIONAL NEURAL NETWORK

A BIT OF HISTORY

Hubel & Wiesel,

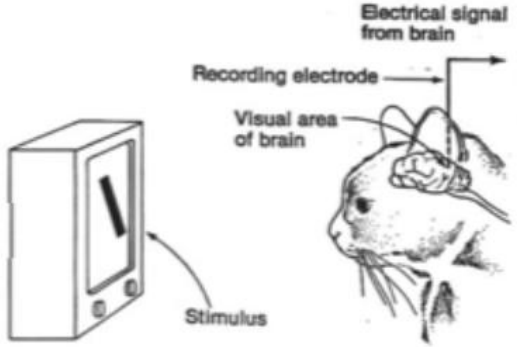
1959

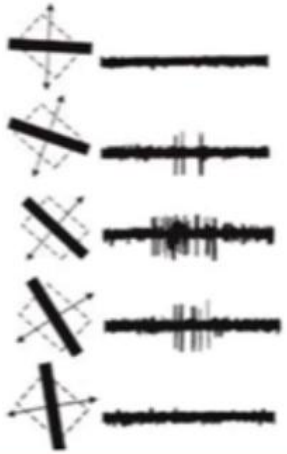
RECEPTIVE FIELDS OF SINGLE NEURONES IN THE CAT'S STRIATE CORTEX

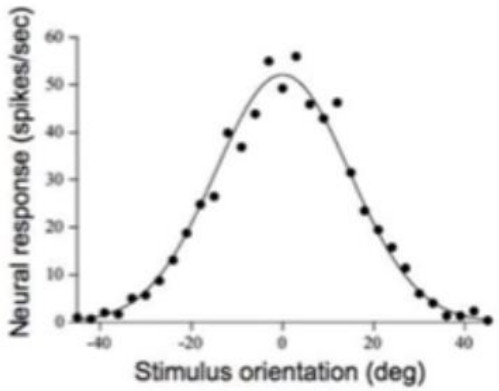
1962

RECEPTIVE FIELDS, BINOCULAR INTERACTION AND FUNCTIONAL ARCHITECTURE IN THE CAT'S VISUAL CORTEX

1968...

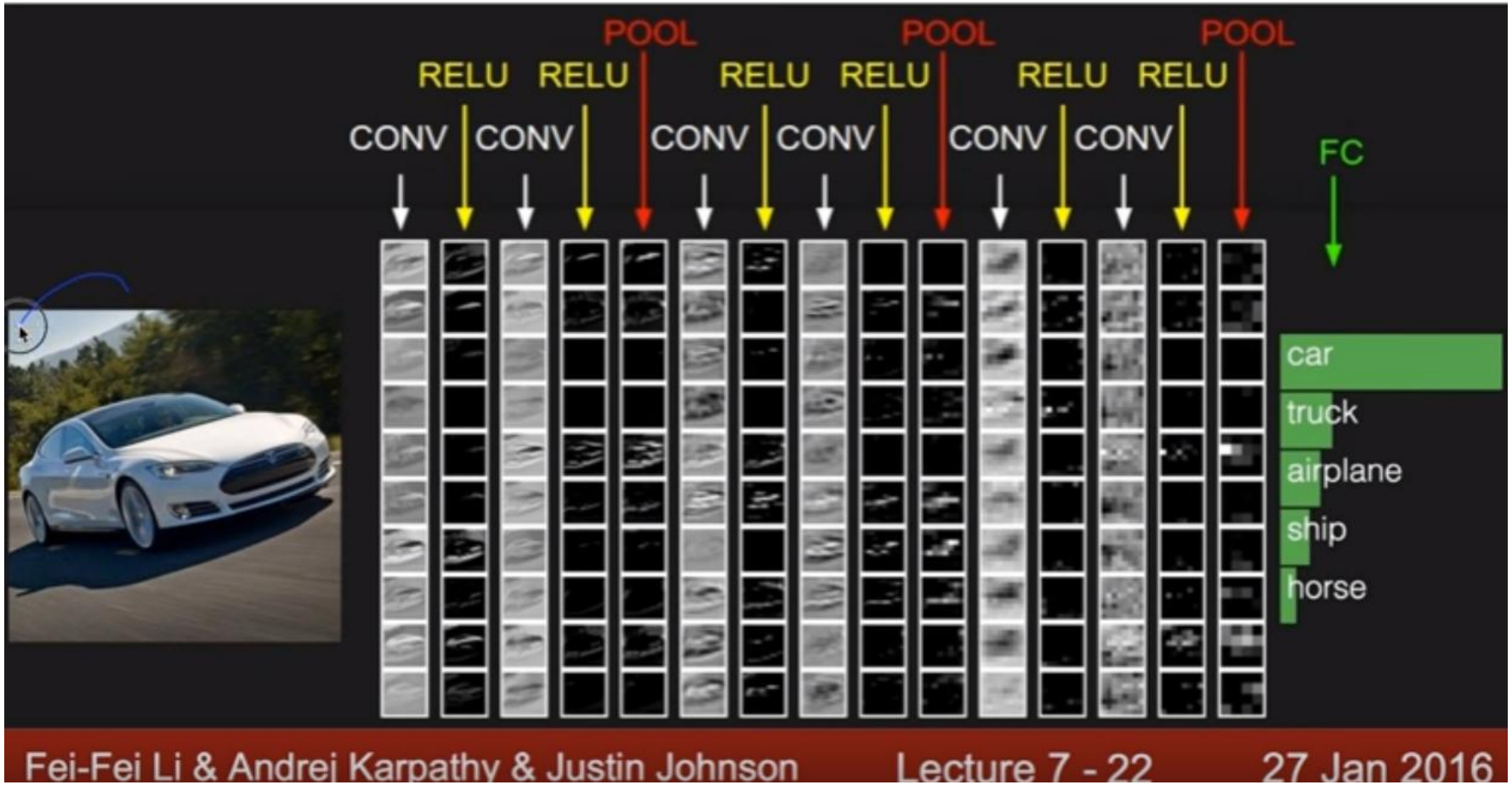






CONVOLUTIONAL NEURAL NETWORK

preview:

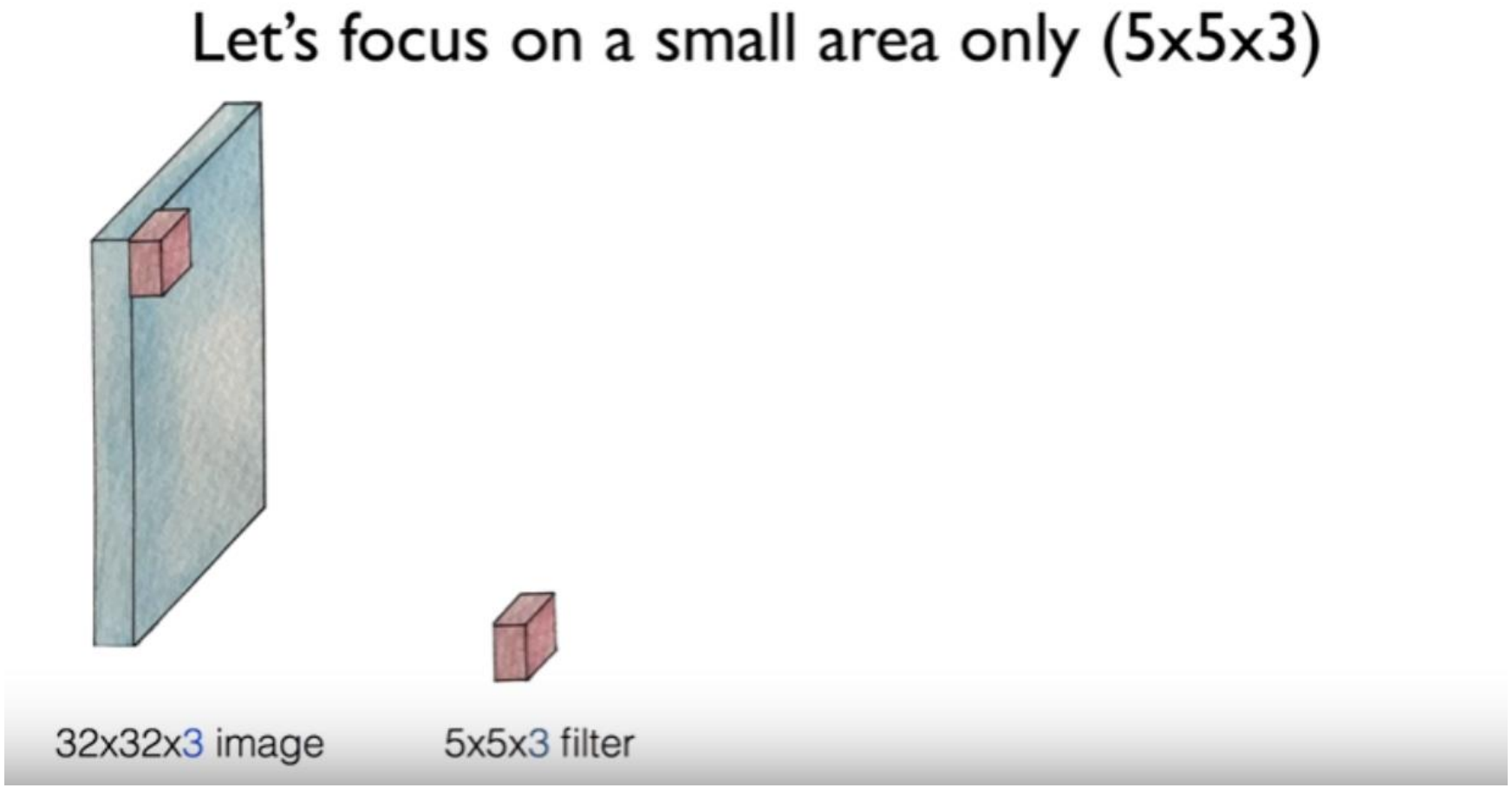


CNN FILTER

Start with an image (width x height x depth)

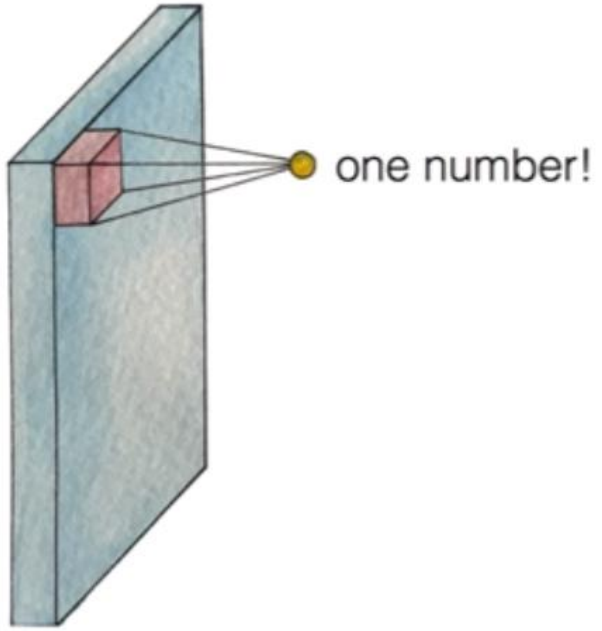
32x32x3 image

CNN FILTER


Let's focus on a small area only (5x5x3)
32x32x3 image
5x5x3 filter

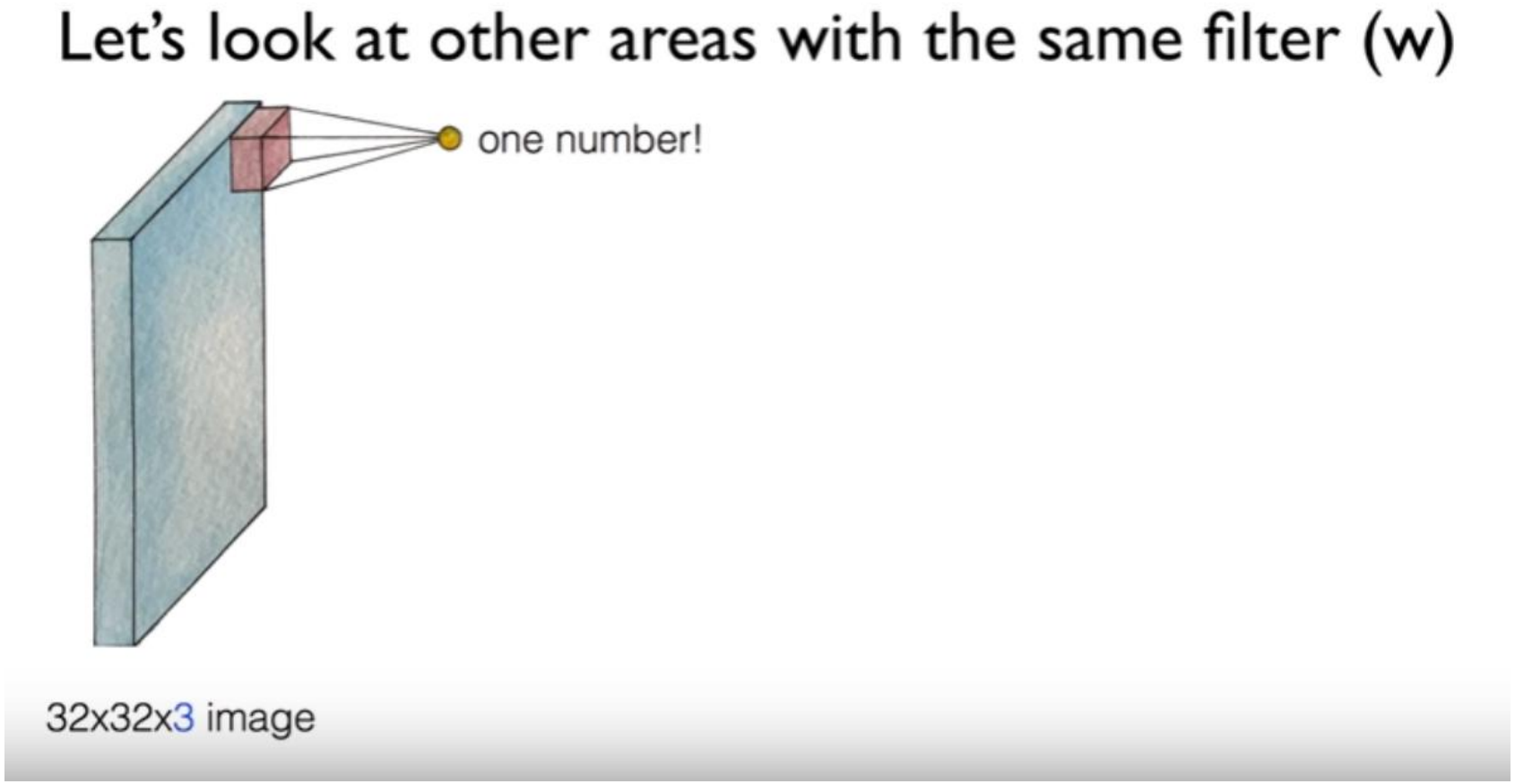
CNN FILTER

Let's look at other areas with the same filter (w)



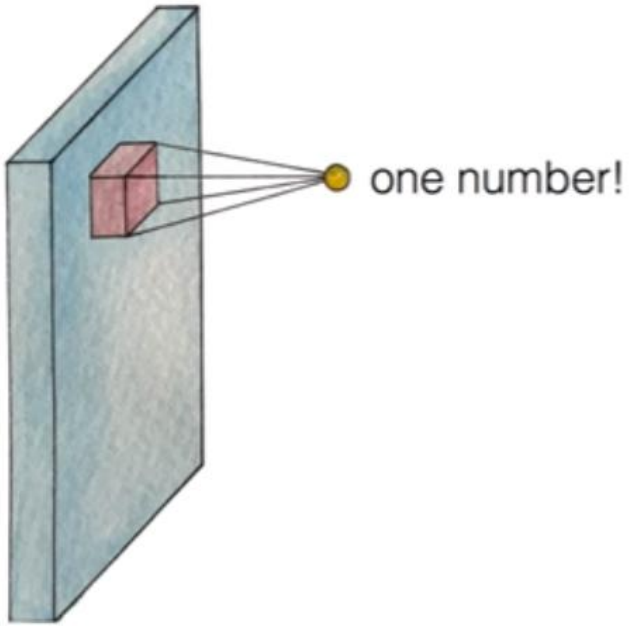
32x32x3 image

CNN FILTER


Let's look at other areas with the same filter (w)
one number!
32x32x3 image

CNN FILTER

Let's look at other areas with the same filter (w)

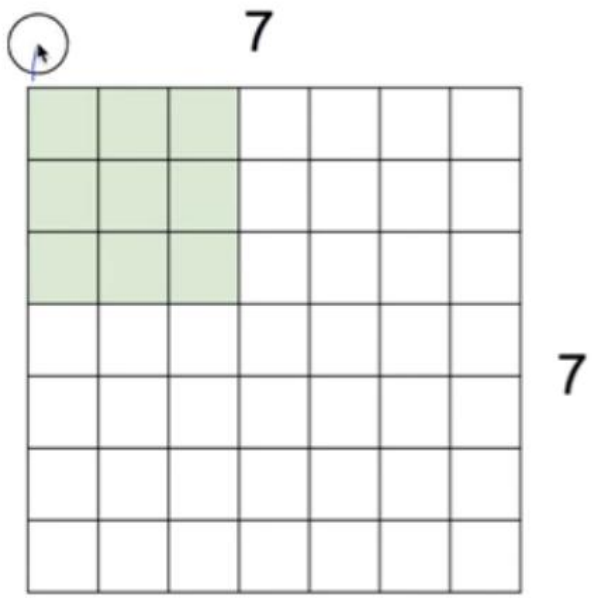


How many numbers can we get?

32x32x3 image

MOVING FILTER

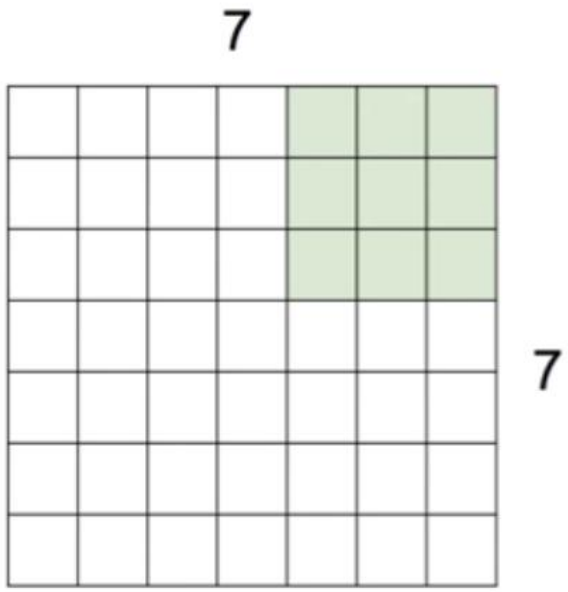
A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter

MOVING FILTER

A closer look at spatial dimensions:

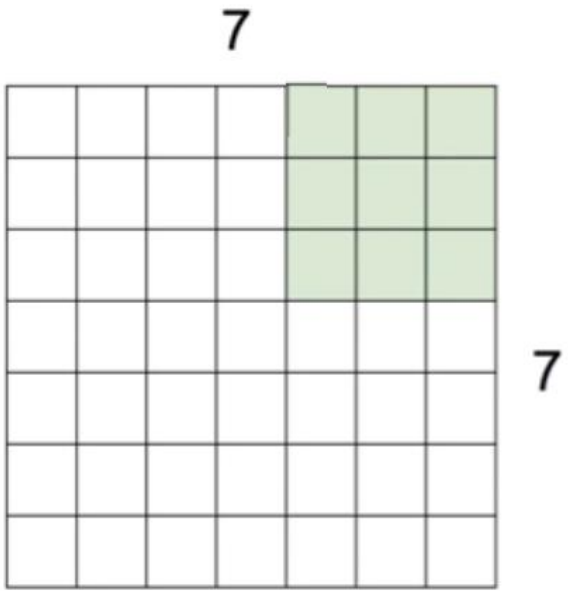


7x7 input (spatially)
assume 3x3 filter

=> 5x5 output

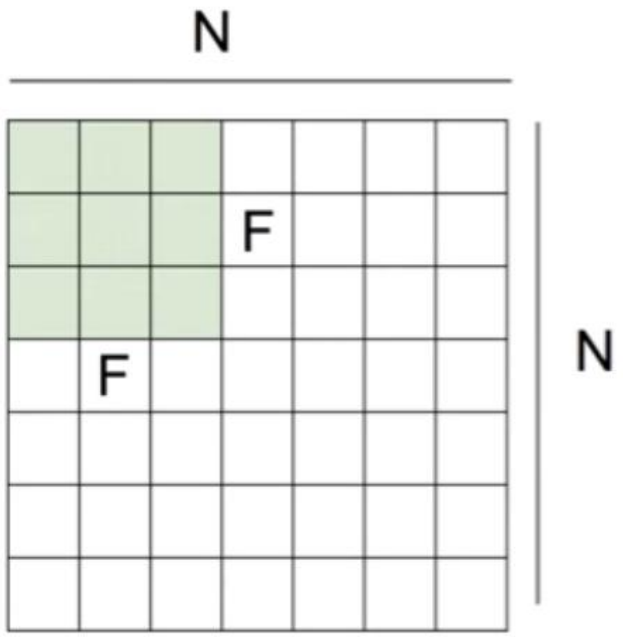
MOVING FILTER

A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter
applied **with stride 2**
=> 3x3 output!

MOVING FILTER

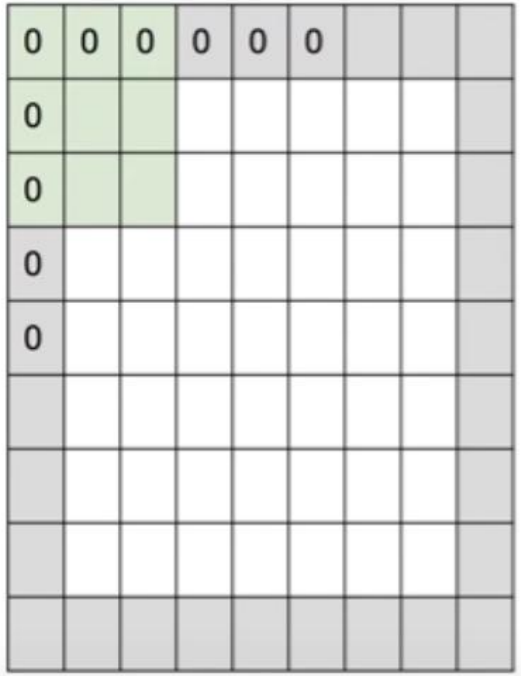


Output size:
(N - F) / stride + 1

e.g. N = 7, F = 3:
stride 1 => (7 - 3)/1 + 1 = 5
stride 2 => (7 - 3)/2 + 1 = 3
stride 3 => (7 - 3)/3 + 1 = 2.33 :\

ZERO PADDINGS

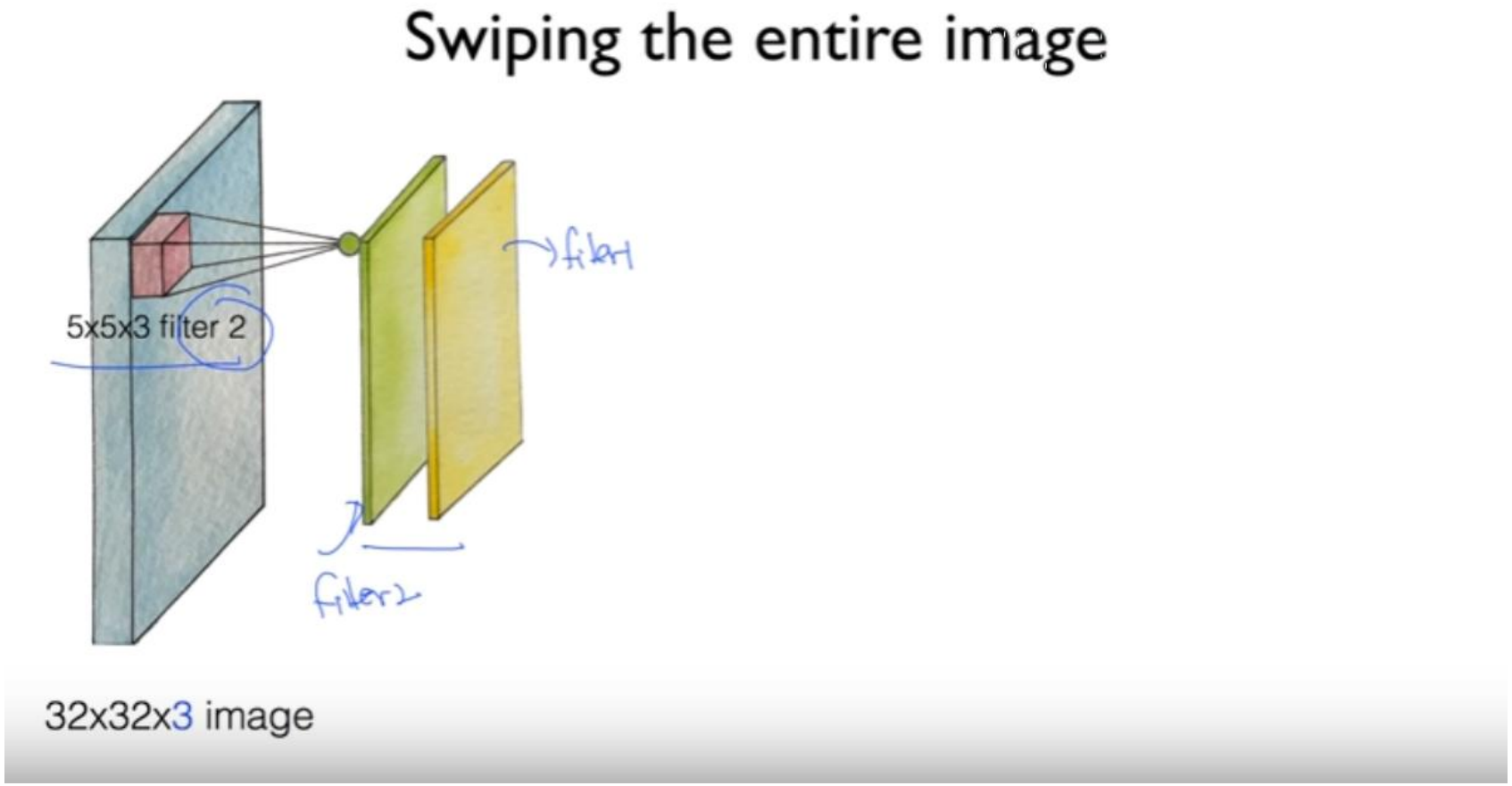
In practice: Common to zero pad the border



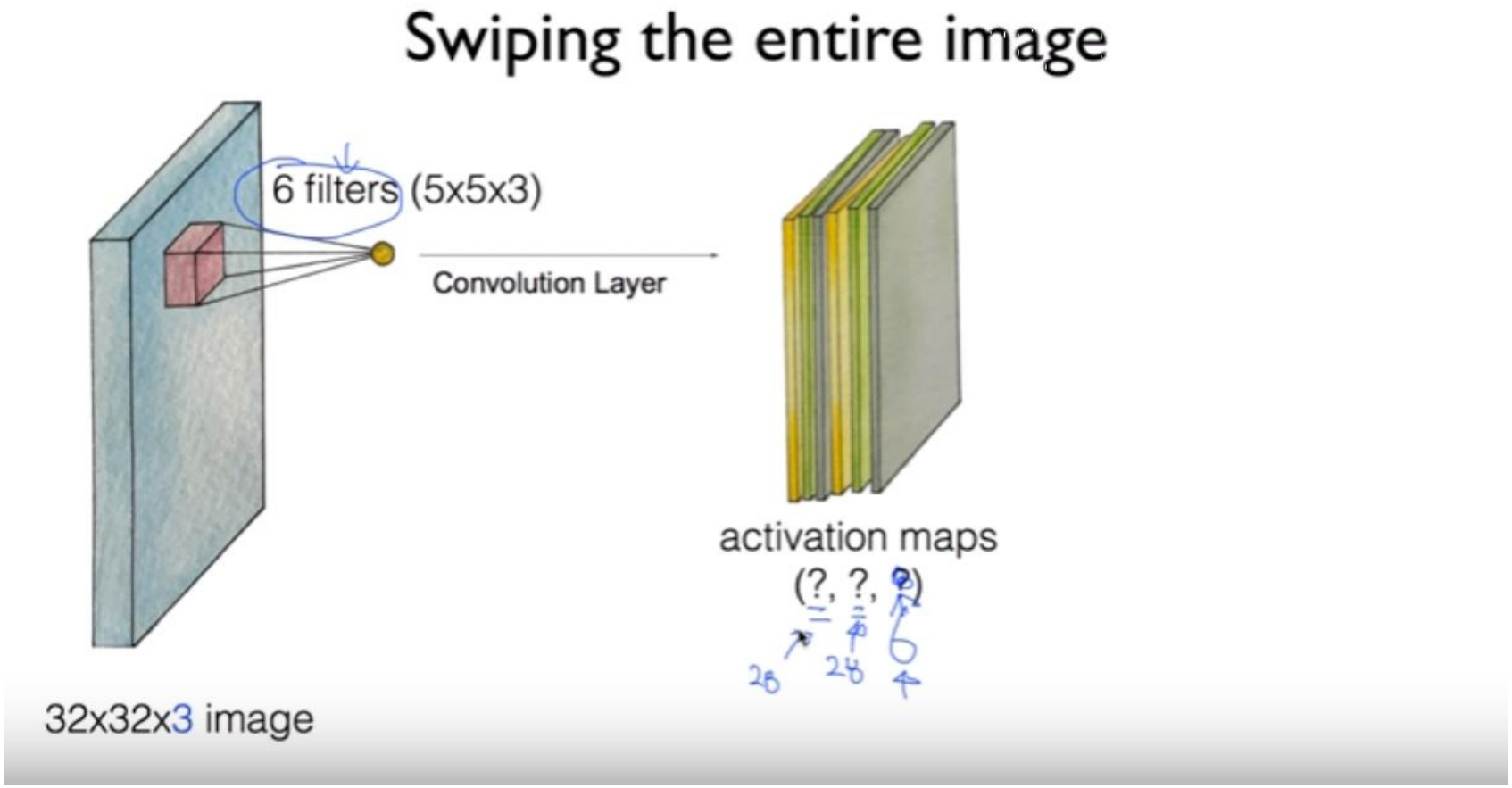
e.g. input 7x7
3x3 filter, applied with **stride 1**
pad with 1 pixel border => what is the output?

7x7 output!
in general, common to see CONV layers with stride 1, filters of size FxF, and zero-padding with (F-1)/2. (will preserve size spatially)
e.g. F = 3 => zero pad with 1
F = 5 => zero pad with 2
F = 7 => zero pad with 3

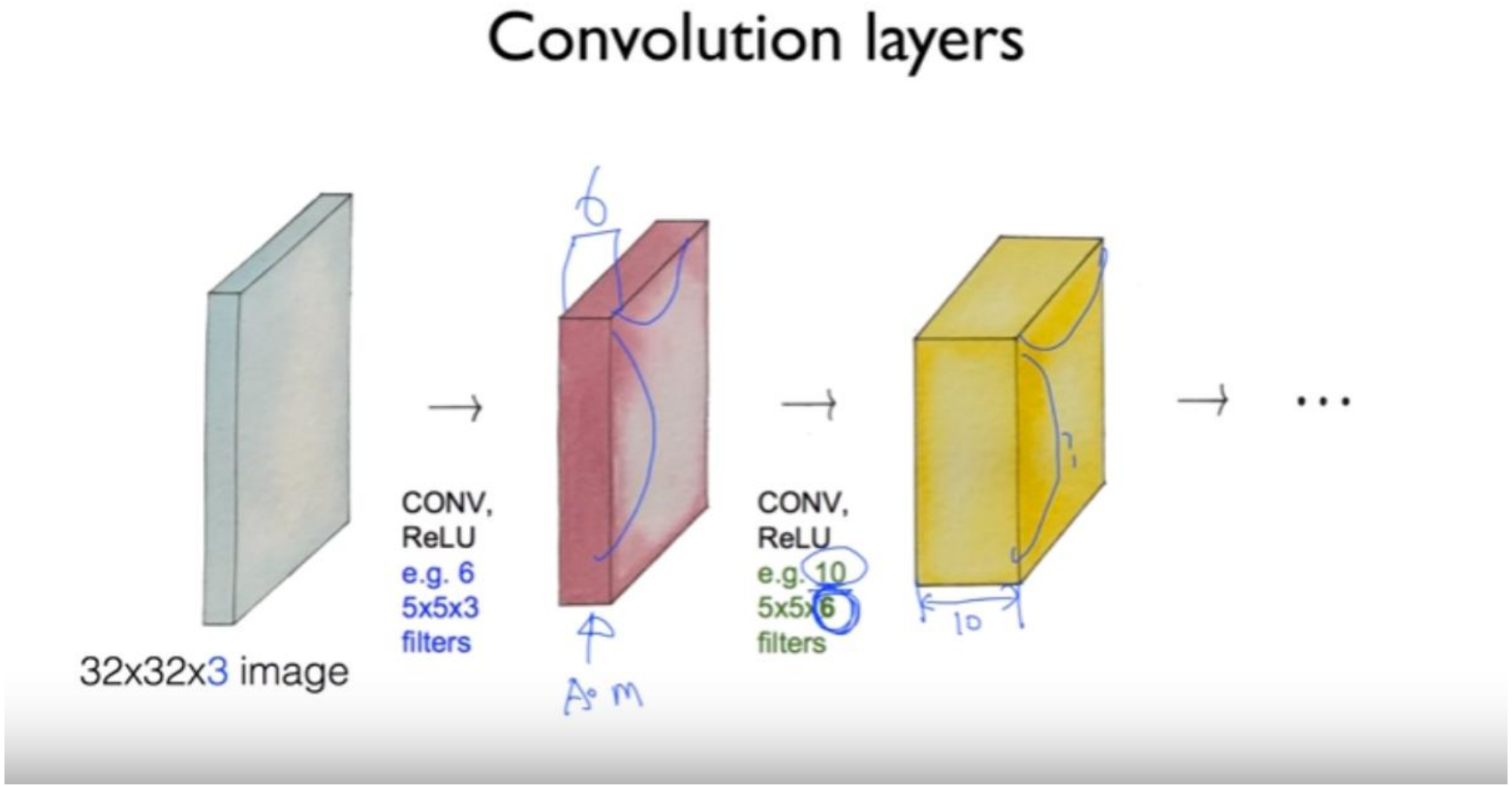
CONVOLUTIONAL LAYERS



CONVOLUTIONAL LAYERS

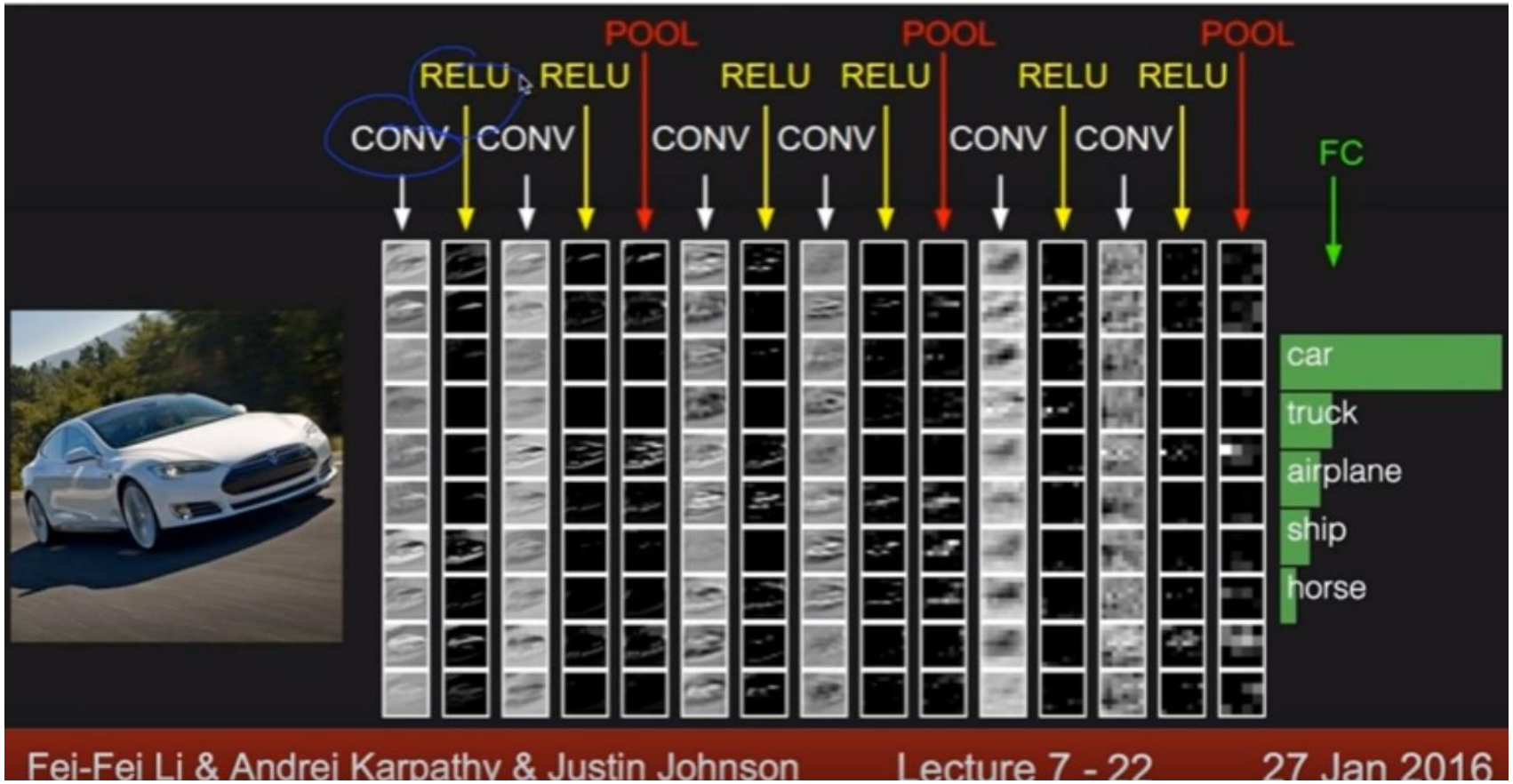


CONVOLUTIONAL LAYERS

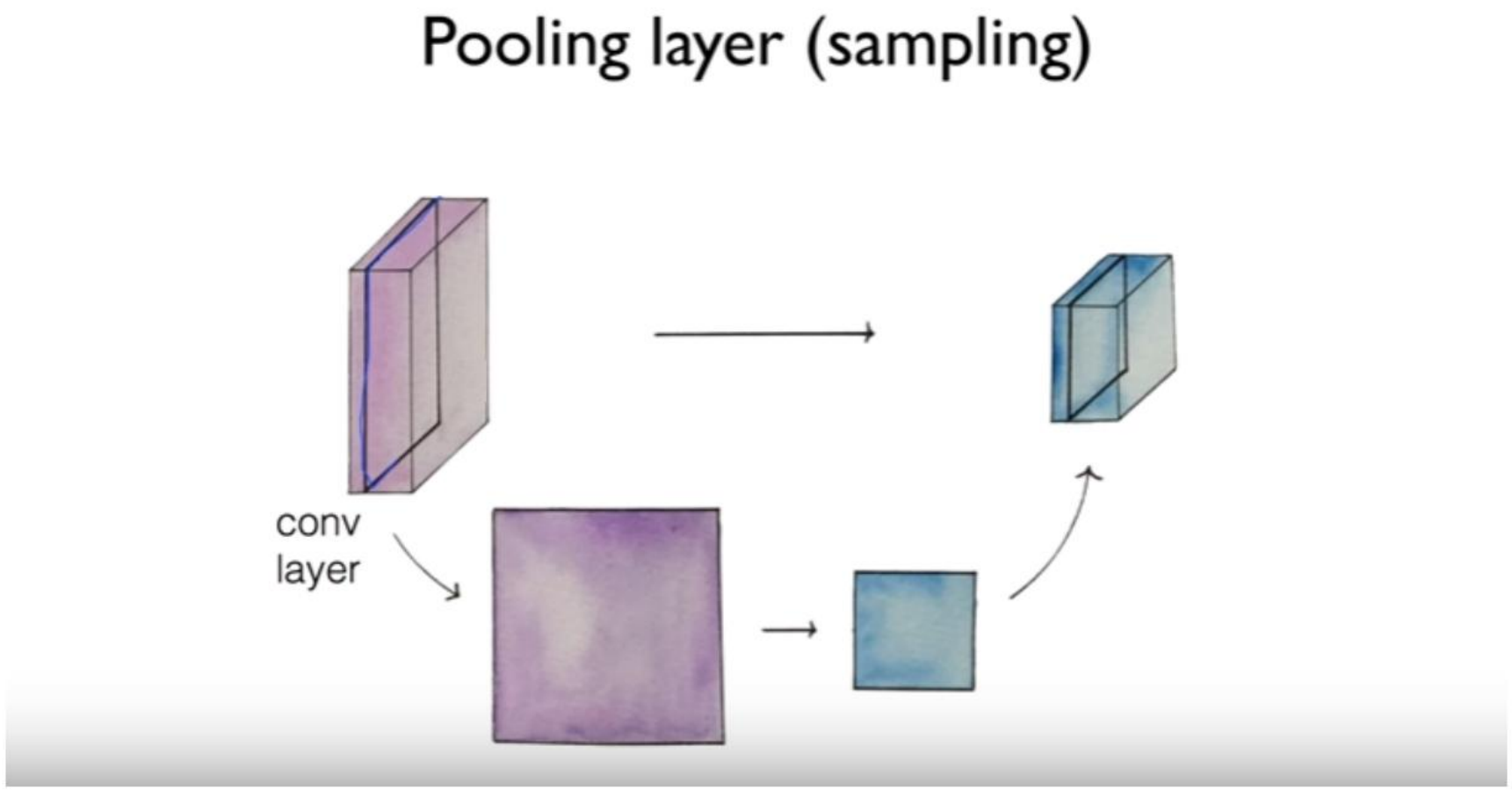


POOLING LAYER

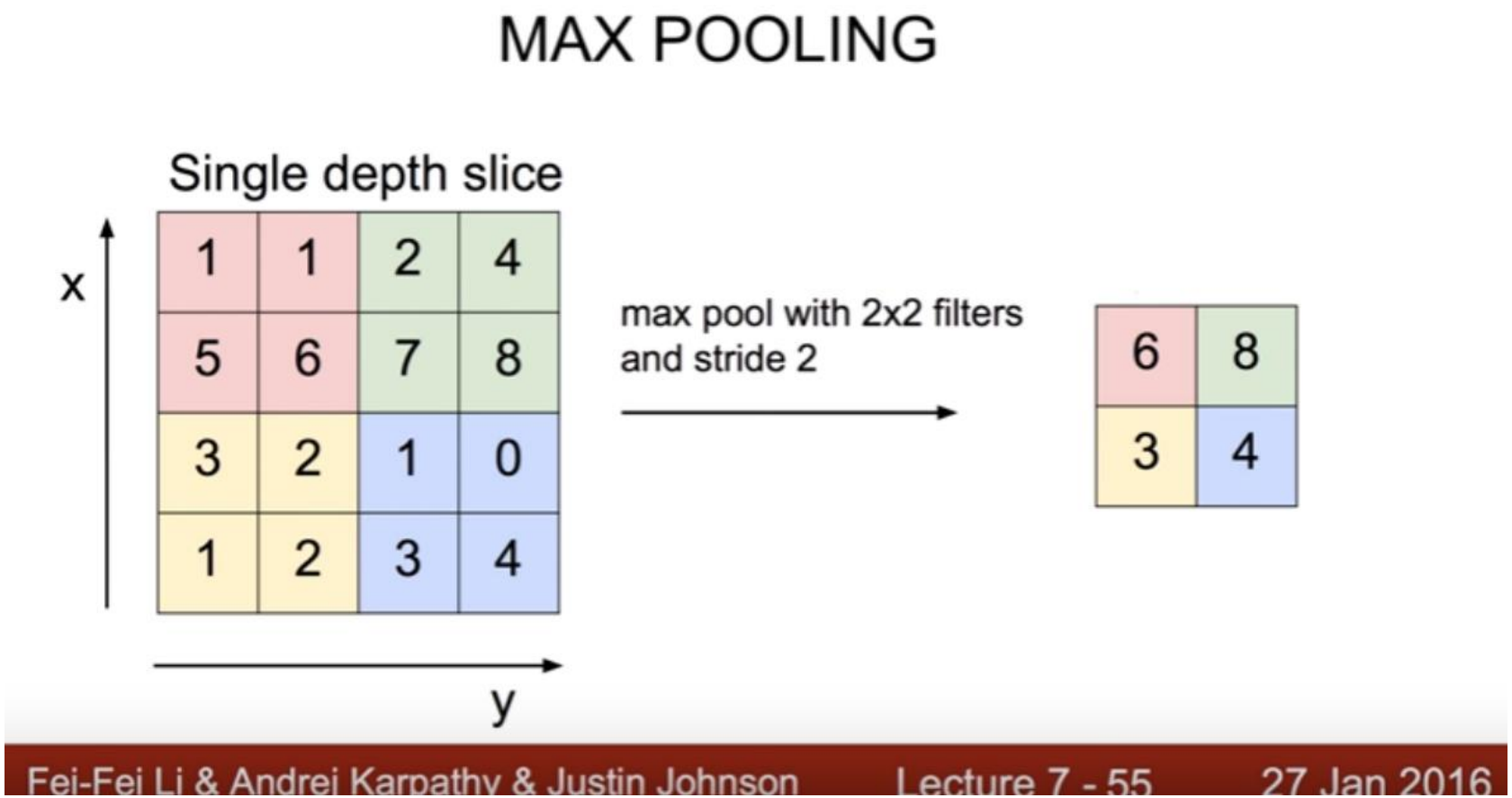
preview:



POOLING LAYER

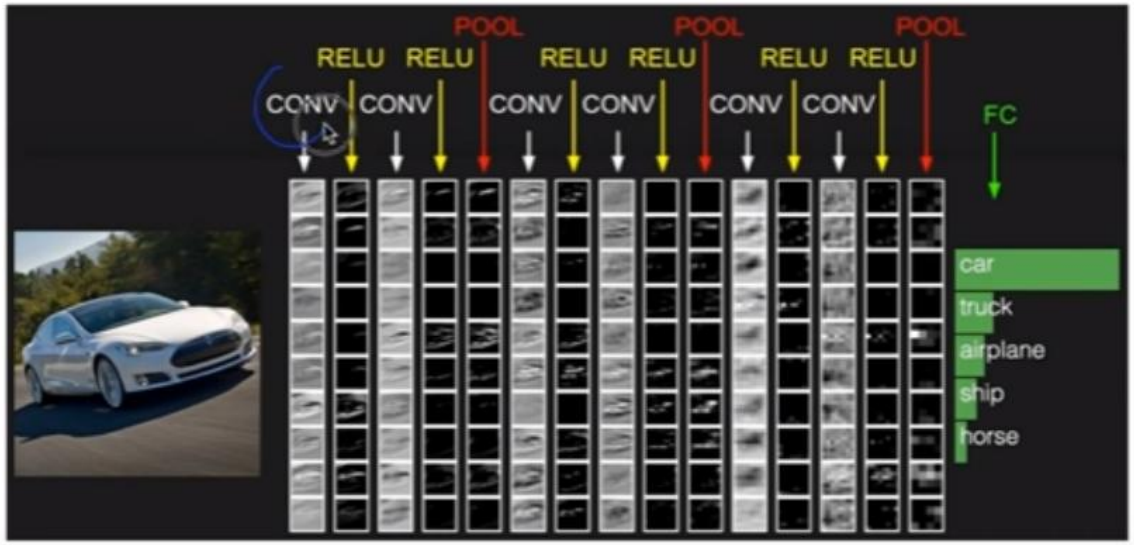


POOLING LAYER



FULLY CONNECTED LAYER

- Contains neurons that connect to the entire input volume, as in ordinary Neural Networks

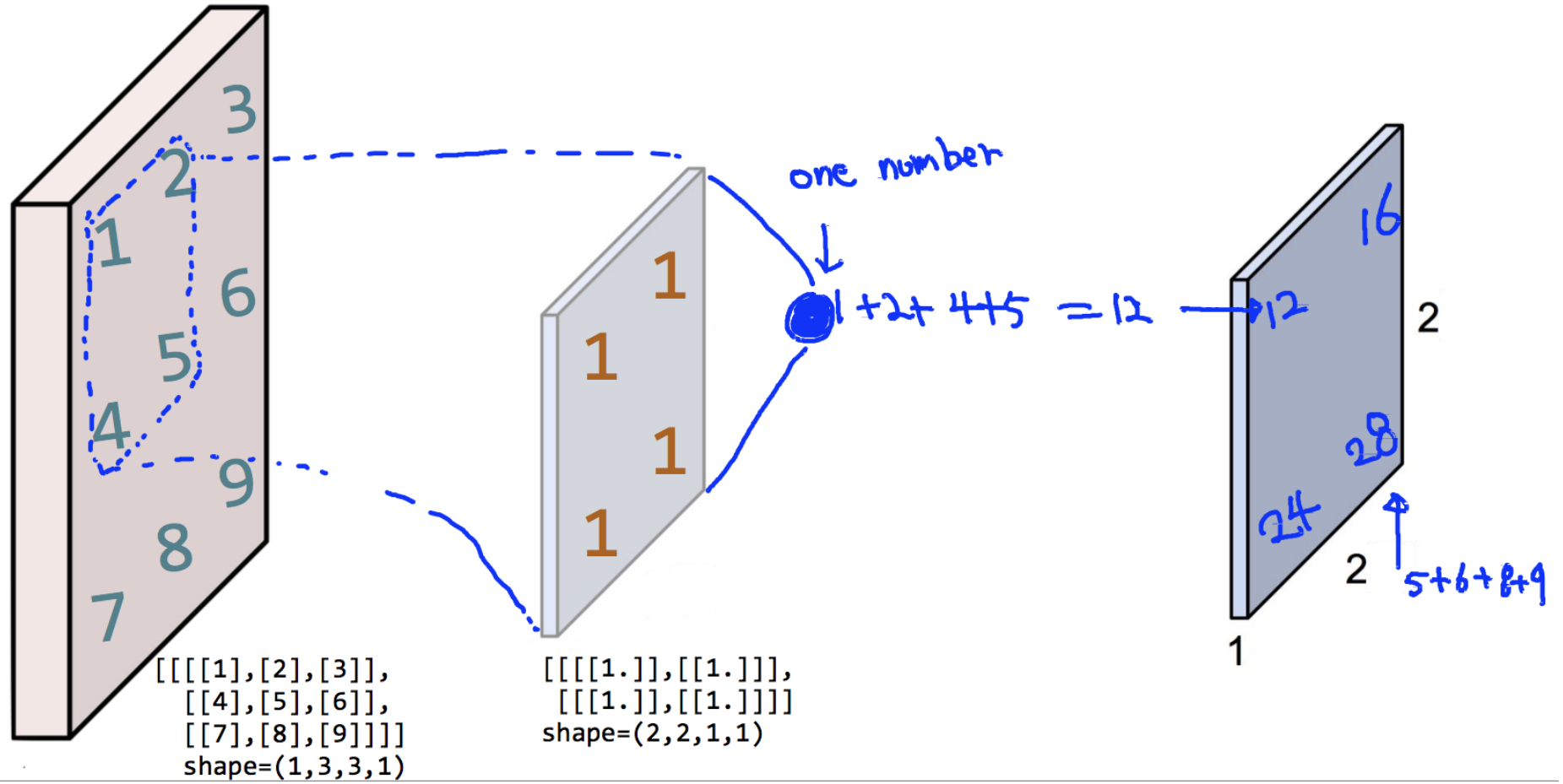


Fei-Fei Li & Andrej Karpathy & Justin Johnson Lecture 7 - 58 27 Jan 2016

http://cs.stanford.edu/people/karpathy/convnetjs/demo/cifar10.html

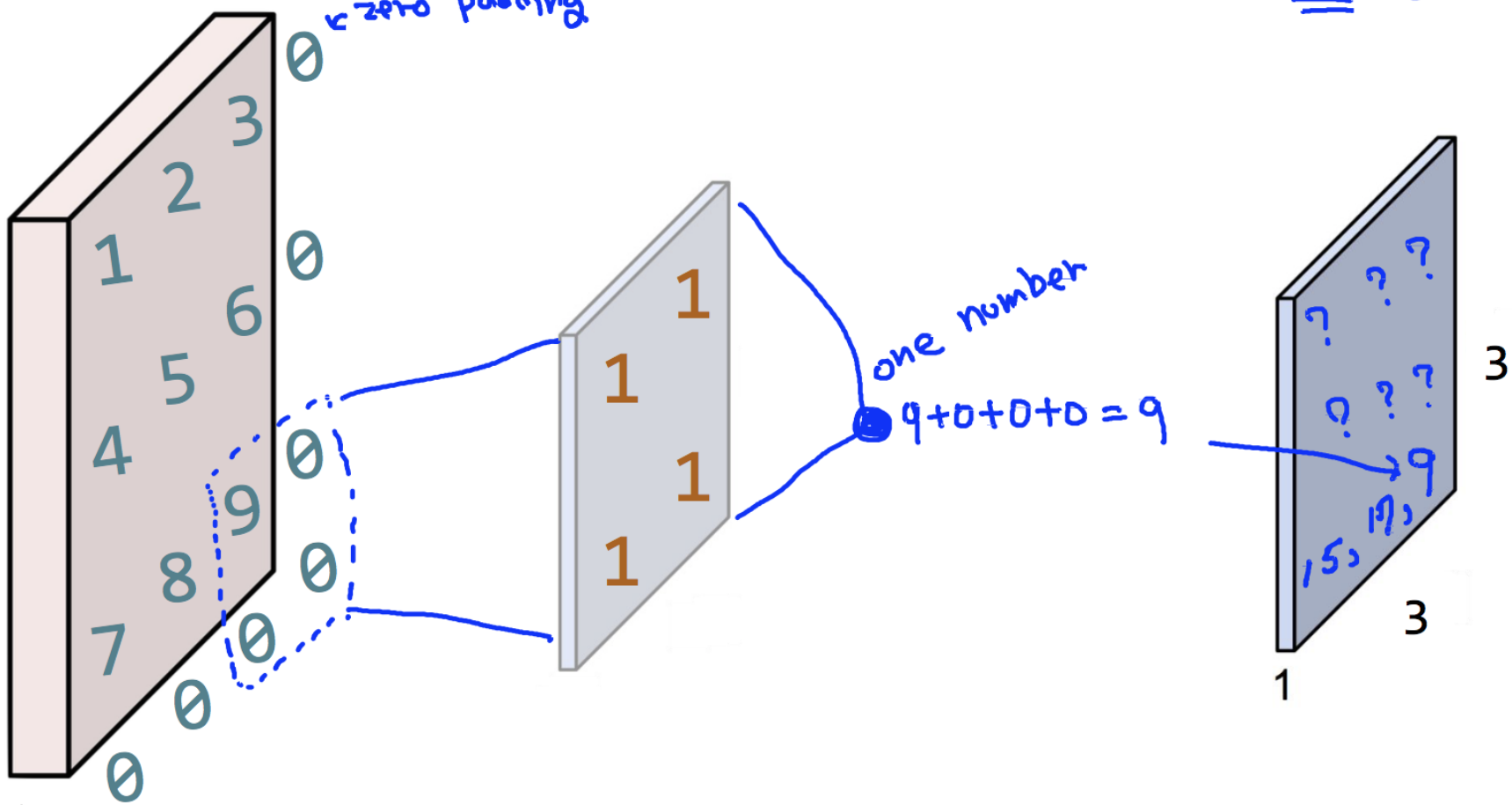
Simple convolution layer

Image: 1,3,3,1 image, Filter: 2,2,1,1, Stride: 1x1, Padding: VALID



Simple convolution layer

Image: 1,3,3,1 image, Filter: 2,2,1,1, Stride: 1x1, Padding: SAME want 3x3



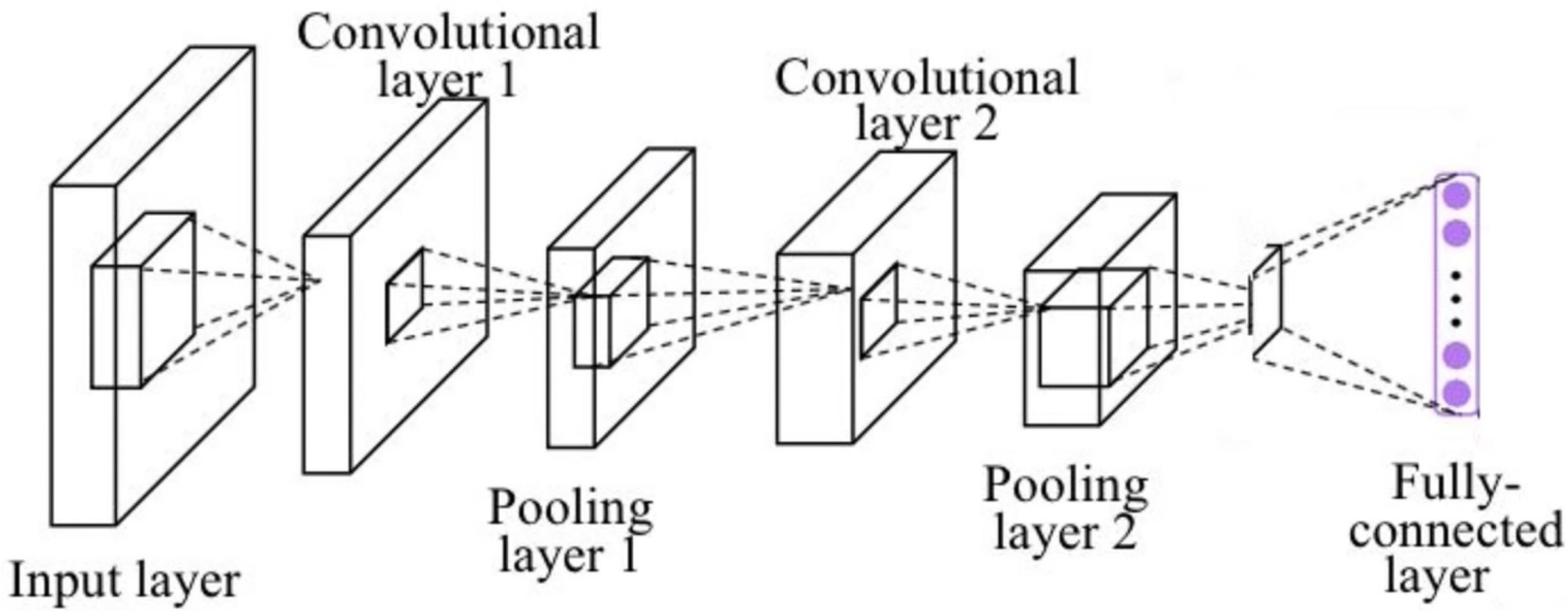
4	3	0
2	1	0
0	0	0

4	3	0
2	1	0
0	0	0

4	3	0
2	1	0
0	0	0

4	3	0
2	1	0
0	0	0

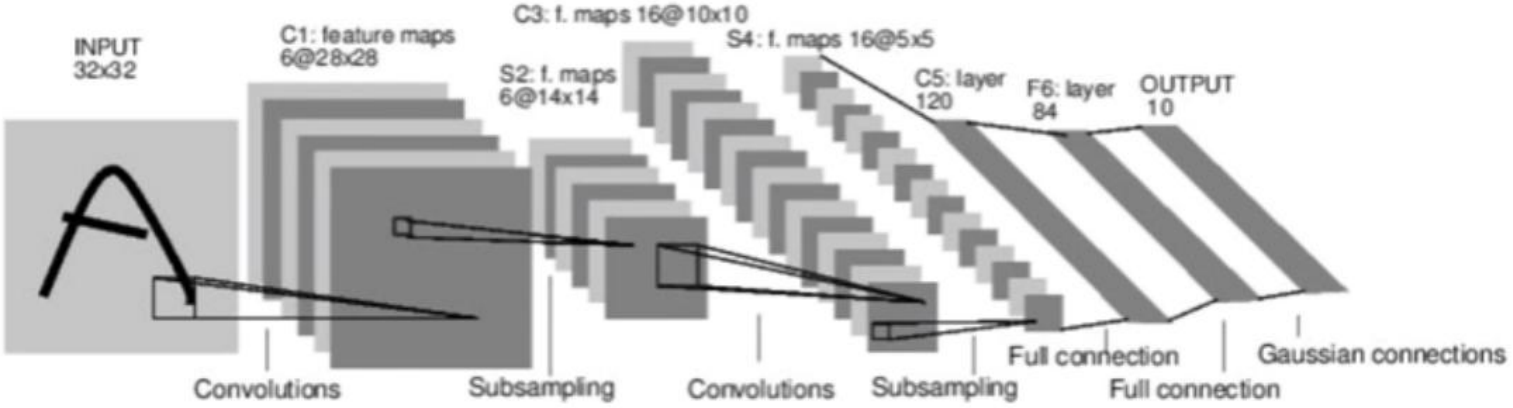
Simple CNN


Convolutional layer 1
Convolutional layer 2
Input layer
Pooling layer 1
Pooling layer 2
Fully-connected layer

CASE STUDY : LENET-5

Case Study: LeNet-5

[LeCun et al., 1998]

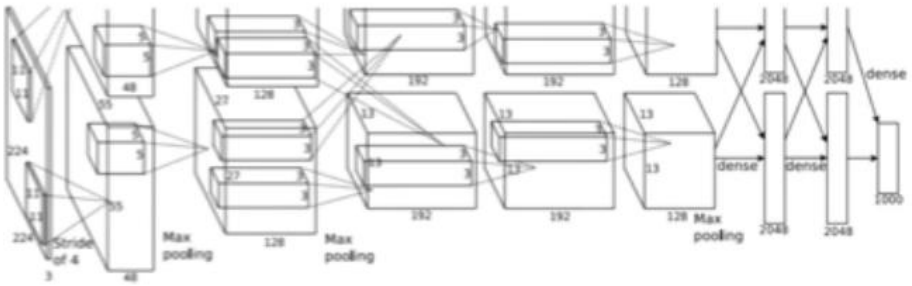


Conv filters were 5x5, applied at stride 1
Subsampling (Pooling) layers were 2x2 applied at stride 2
i.e. architecture is [CONV-POOL-CONV-POOL-CONV-FC]

CASE STUDY : ALEXNET

Case Study: AlexNet

[Krizhevsky et al. 2012]



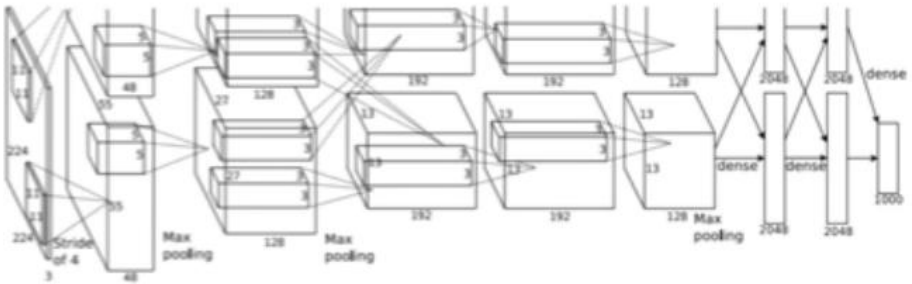
Input: 227x227x3 images

First layer (CONV1): 96 11x11 filters applied at stride 4
=>
Output volume **[55x55x96]**
Parameters: (11*11*3)*96 = **35K**

CASE STUDY : ALEXNET

Case Study: AlexNet

[Krizhevsky et al. 2012]



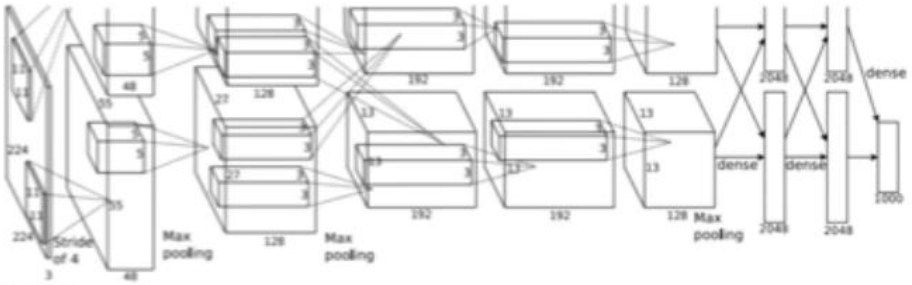
Input: 227x227x3 images
After CONV1: 55x55x96

Second layer (POOL1): 3x3 filters applied at stride 2
Output volume: 27x27x96
Parameters: 0!

CASE STUDY : ALEXNET

Case Study: AlexNet

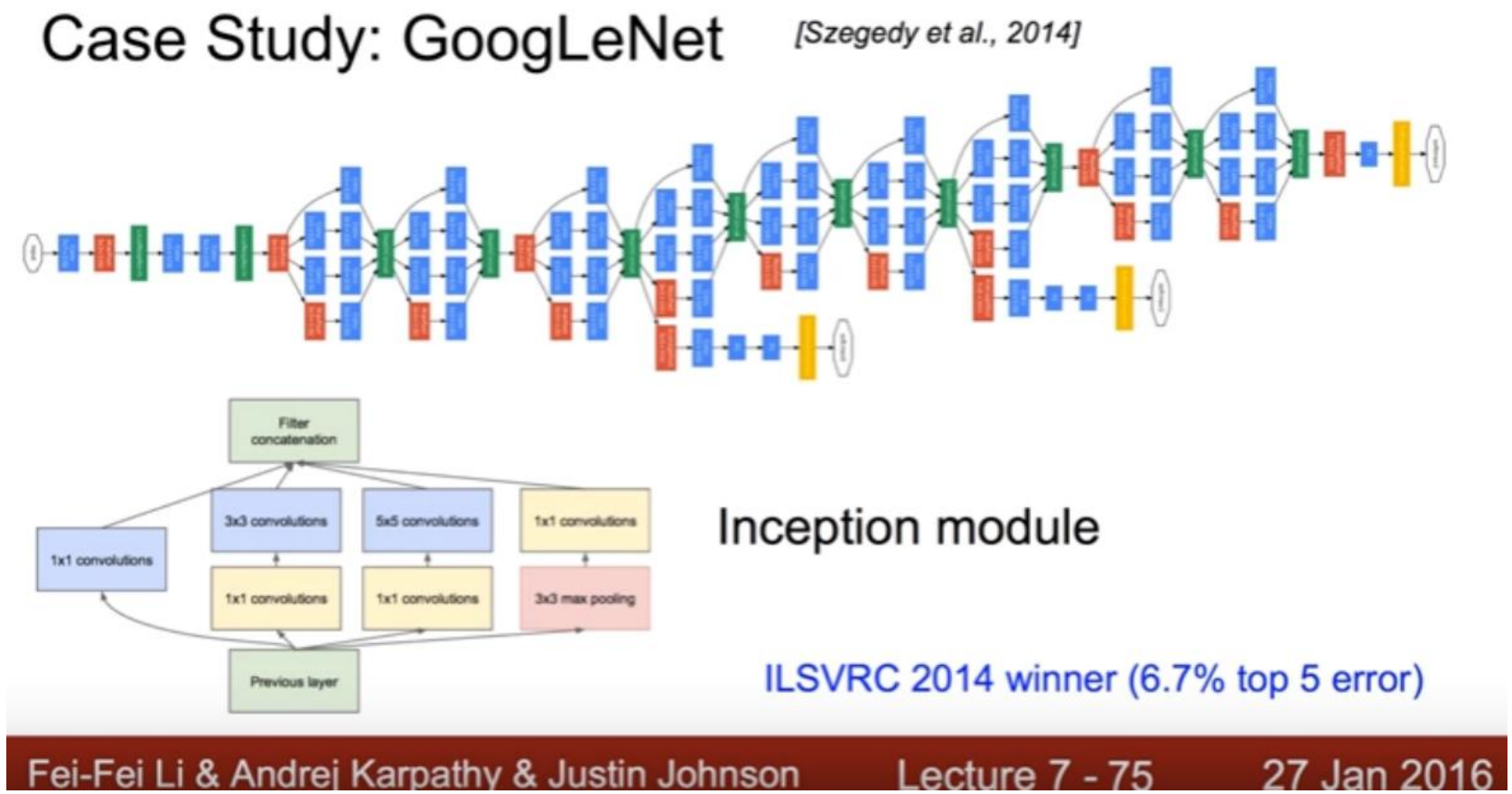
[Krizhevsky et al. 2012]



Full (simplified) AlexNet architecture:
[227x227x3] INPUT
[55x55x96] CONV1: 96 11x11 filters at stride 4, pad 0
[27x27x96] MAX POOL1: 3x3 filters at stride 2
[27x27x96] NORM1: Normalization layer
[27x27x256] CONV2: 256 5x5 filters at stride 1, pad 2
[13x13x256] MAX POOL2: 3x3 filters at stride 2
[13x13x256] NORM2: Normalization layer
[13x13x384] CONV3: 384 3x3 filters at stride 1, pad 1
[13x13x384] CONV4: 384 3x3 filters at stride 1, pad 1
[13x13x256] CONV5: 256 3x3 filters at stride 1, pad 1
[6x6x256] MAX POOL3: 3x3 filters at stride 2
[4096] FC6: 4096 neurons
[4096] FC7: 4096 neurons
[1000] FC8: 1000 neurons (class scores)

Details/Retrospectives:
- first use of ReLU
- used Norm layers (not common anymore)
- heavy data augmentation
- dropout 0.5
- batch size 128
- SGD Momentum 0.9
- Learning rate 1e-2, reduced by 10 manually when val accuracy plateaus
- L2 weight decay 5e-4
- 7 CNN ensemble: 18.2% -> 15.4%

CASE STUDY : GOOGLENET



CASE STUDY : RESNET

Case Study: ResNet *[He et al., 2015]*

ILSVRC 2015 winner (3.6% top 5 error)

Research

MSRA @ ILSVRC & COCO 2015 Competitions

- **1st places in all five main tracks**
  - ImageNet Classification: ***"Ultra-deep"*** (quote Yann) **152-layer nets**
  - ImageNet Detection: **16%** better than 2nd
  - ImageNet Localization: **27%** better than 2nd
  - COCO Detection: **11%** better than 2nd
  - COCO Segmentation: **12%** better than 2nd

*improvements are relative numbers

ICCV15

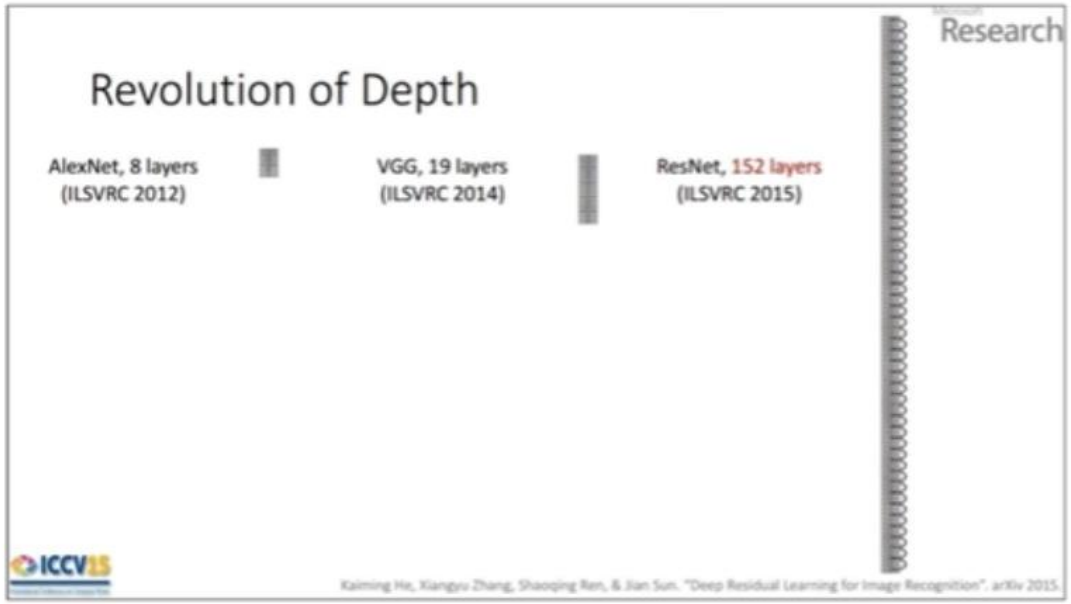
Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Deep Residual Learning for Image Recognition". arXiv 2015.

Slide from Kaiming He's recent presentation https://www.youtube.com/watch?v=1PGLj-uKT1w

CASE STUDY : RESNET

Case Study: ResNet *[He et al., 2015]*

ILSVRC 2015 winner (3.6% top 5 error)

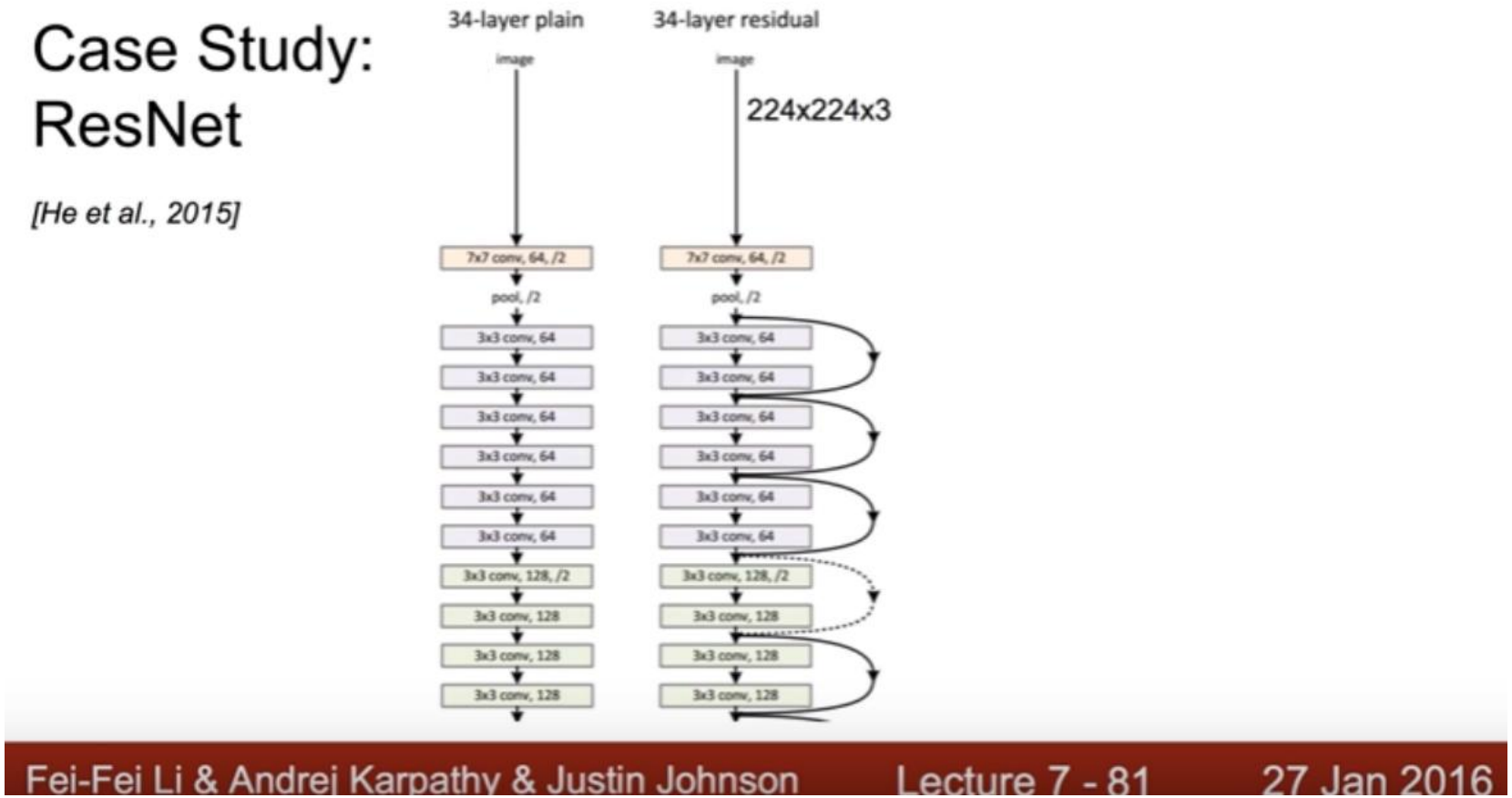


2-3 weeks of training on 8 GPU machine

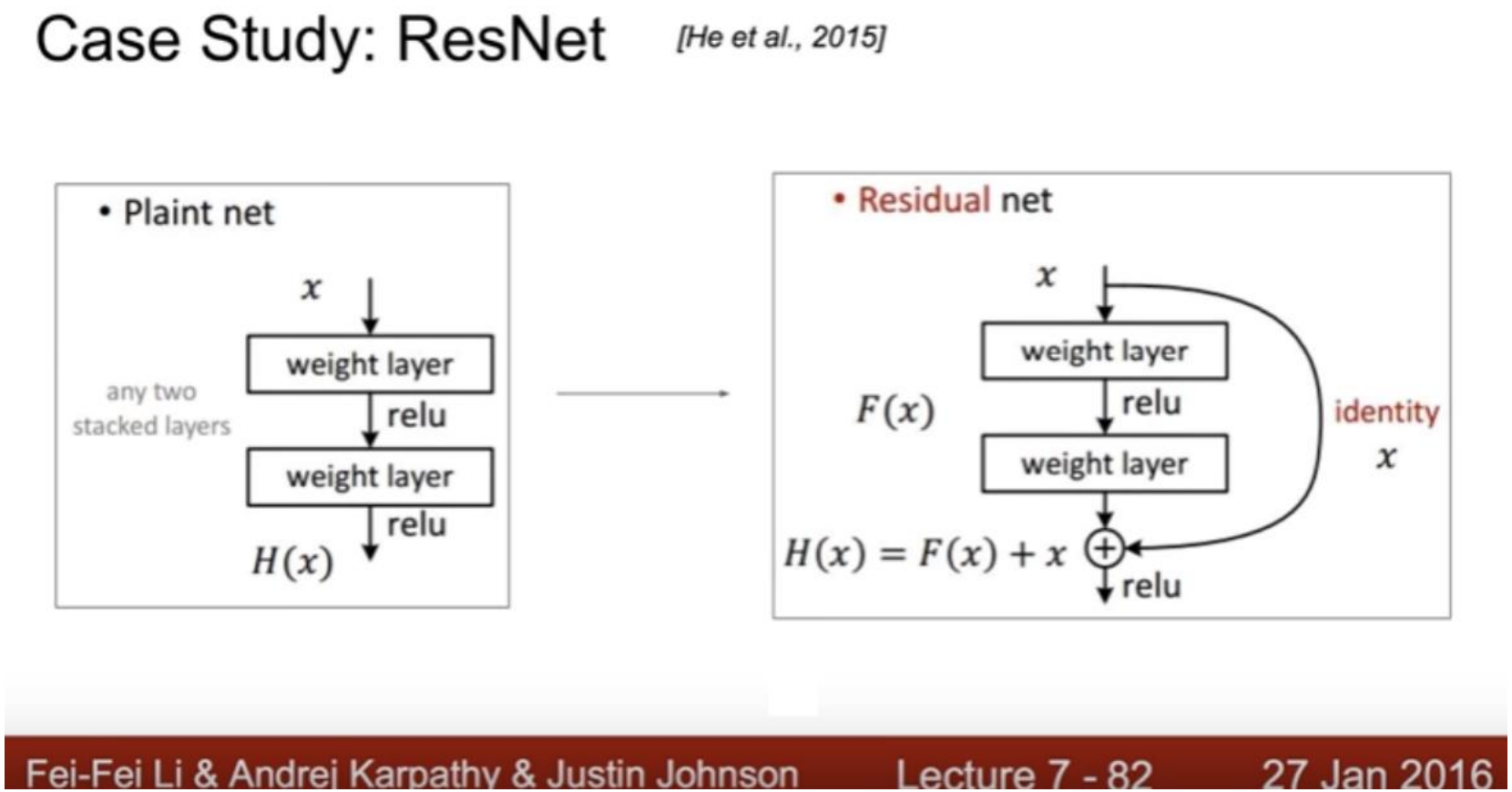
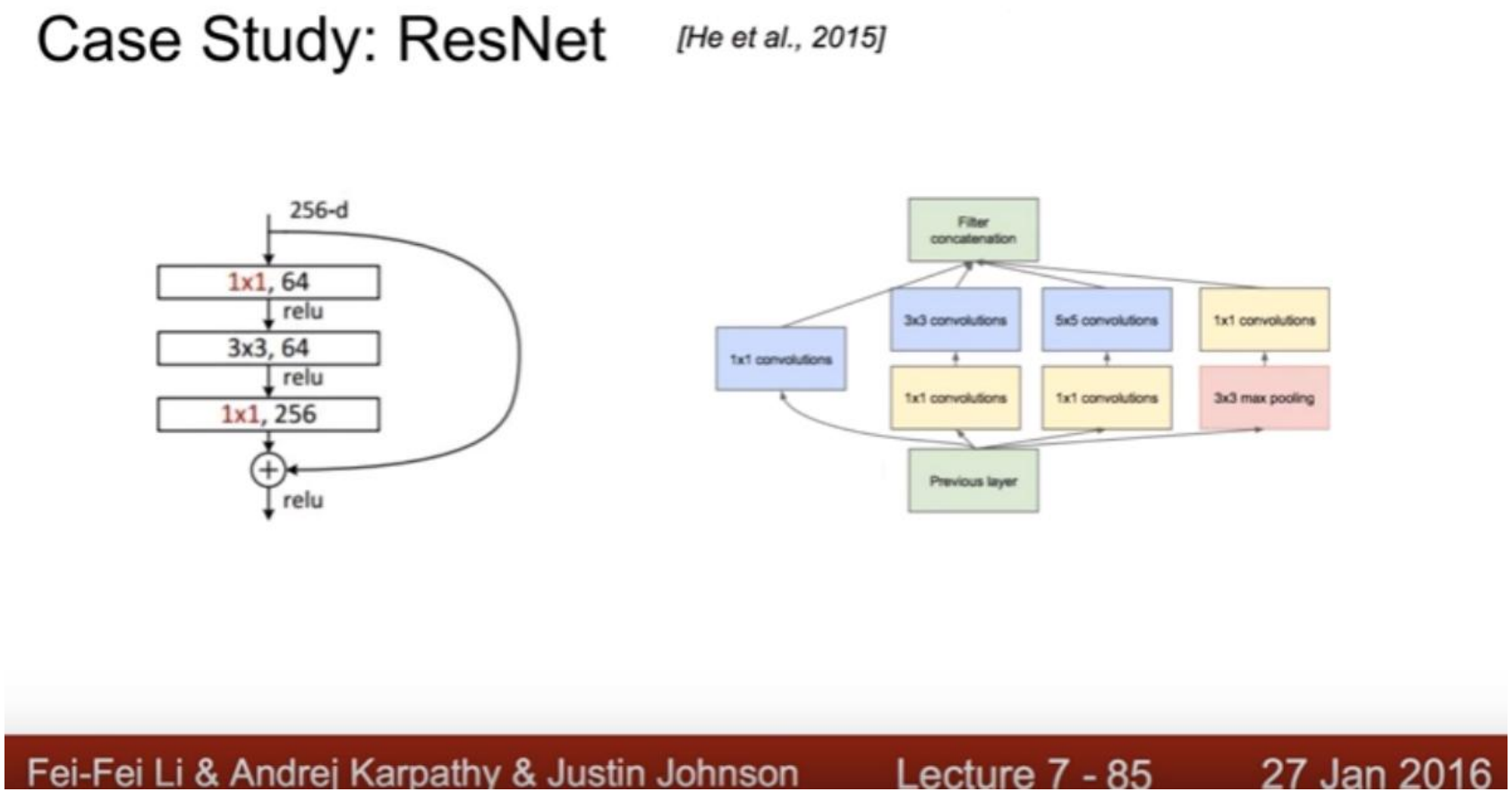
at runtime: faster than a VGGNet! (even though it has 8x more layers)

(slide from Kaiming He's recent presentation)

CASE STUDY : RESNET



CASE STUDY : RESNET



CASE STUDY : RESNET